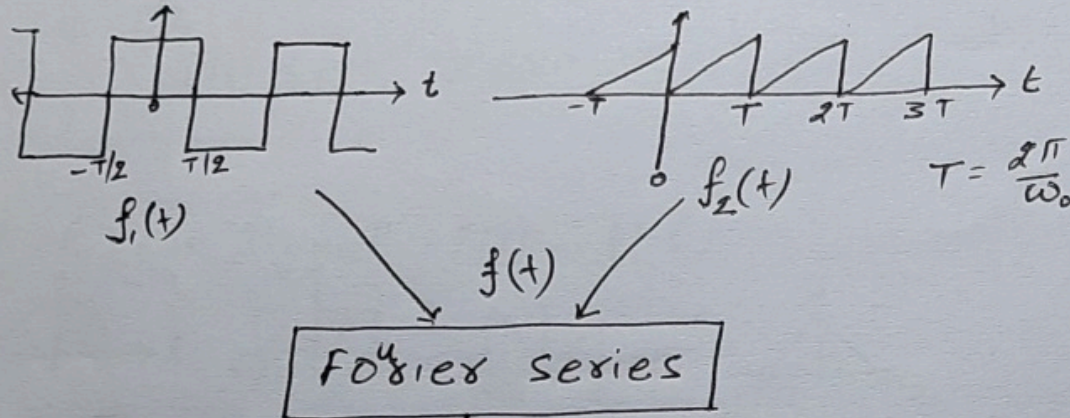


FOURIER SERIES

Concept:

Any Periodic signal with finite Period



Real Fourier Series $\downarrow$ Trigonometric Fourier Series

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t)$$

$\downarrow$ DC Part $\downarrow$ Oscillating Part

* Any Periodic signal ($T = \frac{2\pi}{\omega_0}$) can be transformed into sinusoidal function along with constant term from Fourier Series.

* Fourier Series valid only for Periodic signal.

where a_0, a_n, b_n Trigonometric Fourier Co-efficient.

$$a_0 = \frac{1}{T} \int_{-T/2}^{T/2} f(t) dt \quad \rightarrow \text{Average value.}$$

$$a_n = \frac{2}{T} \int_{-T/2}^{T/2} f(t) \cos n\omega_0 t dt$$

$$b_n = \frac{2}{T} \int_{-T/2}^{T/2} f(t) \sin n\omega_0 t dt$$

Why Fourier Series

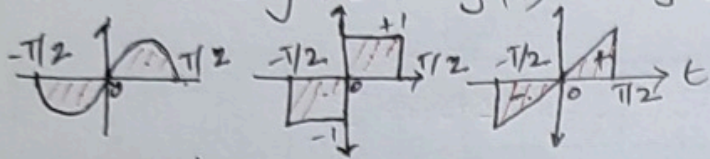
- * Any Periodic fln $\rightarrow$ Sinusoidal fln
- * Easily Represent in mathematical form.
- * Solution for sinusoidal fln quite Convenient

* Differential & integral $e^{j\omega t}$ quite simple.

* F.S is unique for every individual Periodic fln.
 $f_1(t)$ $f_2(t)$

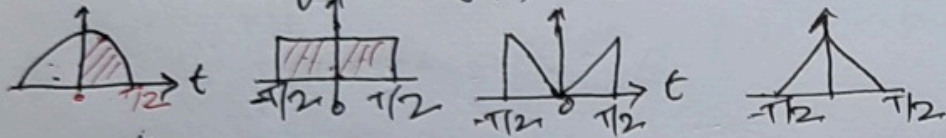
Some Basic Concept

i) odd signal $f(t) = -f(-t)$



$$\int_{-T/2}^{T/2} f(t) dt = \text{zero}$$

ii) Even signal $f(t) = f(-t)$



$$\int_{-T/2}^{T/2} f(t) dt = 2 \int_0^{T/2} f(t) dt$$

$$iii) \int_0^T \cos t dt = \left[\sin t \right]_0^T = \boxed{\sin T - 0}$$

$$iv) \int_0^T \cos 2t dt = \left[\frac{\sin 2t}{2} \right]_0^T = \boxed{\frac{\sin 2T}{2}}$$

$$v) \int_0^T \sin t dt = \left[-\cos t \right]_0^T = -\cos T - (-\cos 0) = \boxed{-\cos T + 1}$$

$$vi) \int_0^T \sin 2t dt = \left[-\frac{\cos 2t}{2} \right]_0^T = -\frac{1}{2} \left[\cos 2T - 1 \right]$$

e.g. $\int_{-T/2}^{T/2} \sin u \omega_0 t dt = P.$ odd

$$\left[-\frac{\cos u \omega_0 t}{u \omega_0} \right]_{-T/2}^{T/2}$$

$$= \frac{1}{u \omega_0} \left[\cos u \omega_0 \frac{T}{2} - \left(\cos u \omega_0 \left(-\frac{T}{2} \right) \right) \right]$$

$$= \frac{1}{u \omega_0} \left[\cos u \omega_0 \frac{T}{2} - \cos u \omega_0 \frac{T}{2} \right] = 0$$

Note:-

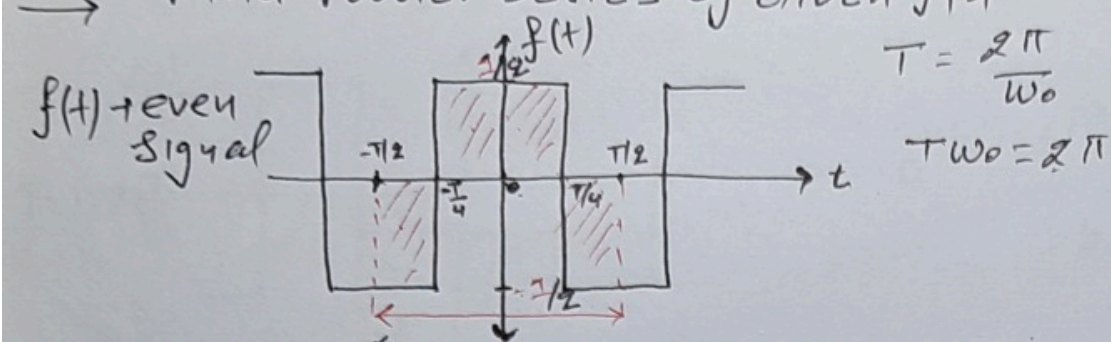
i) $\int_{-T/2}^{T/2} \sin t \times \text{square wave} dt = \text{even}$
odd $\times$ odd = even

ii) $\int_{-T/2}^{T/2} \cos t \times \text{square wave} dt = \text{odd}$
odd $\times$ even = odd

iii) $\int_{-T/2}^{T/2} \sin t \times \text{triangular wave} dt = \text{even}$
even $\times$ even = even

Problem on Fourier Series

Q-1 Find Fourier series of Given $f(t)$



$$f(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos n\omega_0 t + b_n \sin n\omega_0 t$$

$$a_0 = \frac{1}{T} \int_{-T/2}^{T/2} f(t) \cdot dt = \frac{0}{T} = 0 \quad (\text{No DC part})$$

$$a_n = \frac{2}{T} \int_{-T/2}^{T/2} f(t) \cdot \cos n\omega_0 t \cdot dt$$

$$= \frac{4}{T} \int_0^{T/4} f(t) \cdot \cos n\omega_0 t \cdot dt$$

$$= \frac{4}{T} \left[\int_0^{T/4} \frac{1}{2} \cos n\omega_0 t \cdot dt + \int_{T/4}^{T/2} -\frac{1}{2} \cos n\omega_0 t \cdot dt \right]$$

$$= \frac{4}{T} \left[\frac{1}{2} \left| \frac{\sin n\omega_0 t}{n\omega_0} \right|_0^{T/4} - \frac{1}{2} \left| \frac{\sin n\omega_0 t}{n\omega_0} \right|_{T/4}^{T/2} \right]$$

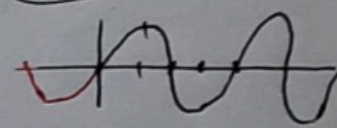
$$a_n = \frac{4}{T} \left[\frac{1}{2} \left(\frac{\sin n\omega_0 T/4}{n\omega_0} - 0 \right) - \frac{1}{2} \left(\frac{\sin n\omega_0 T/2}{n\omega_0} - \frac{\sin n\omega_0 T/4}{n\omega_0} \right) \right]$$

$$= \frac{2}{T} \left[\frac{\sin n\omega_0 T/4}{n\omega_0} - \frac{\sin n\omega_0 T/2}{n\omega_0} + \frac{\sin n\omega_0 T/4}{n\omega_0} \right]$$

$$= \frac{2}{T} \left[2 \frac{\sin n\omega_0 T/4}{n\omega_0} - \frac{\sin n\omega_0 T/2}{n\omega_0} \right]$$

$$= \frac{2}{T\omega_0 n} \left[2 \sin n\pi/2 - \sin n\pi \right] \rightarrow 0$$

$$= \frac{1}{n\pi} \left[2 \sin n\pi/2 \right]$$



$$a_n = \frac{2}{n\pi} \sin n\pi/2$$

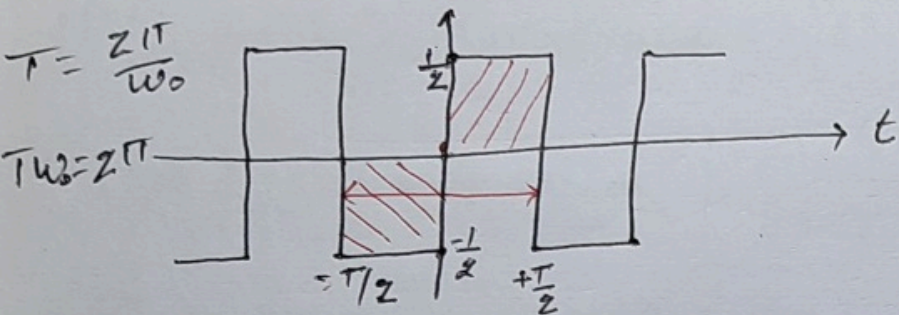
$$b_n = \frac{2}{T} \int_{-T/2}^{T/2} \underbrace{f(t)}_{\text{even}} \cdot \underbrace{\sin n\omega_0 t}_{\text{odd}} \cdot dt = 0$$

$$f(t) = \sum_{n=1}^{\infty} \left(\frac{2}{n\pi} \sin n\frac{\pi}{2} \right) \cos n\omega_0 t$$

✓

Basic Problem on F.S

Q.2 find FS for given $f(t)$



$$f(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos n\omega_0 t + b_n \sin n\omega_0 t$$

$$a_0 = \frac{1}{T} \int_{-T/2}^{T/2} f(t) \cdot dt = \frac{1}{T} \left[\int_{-T/2}^0 \frac{1}{2} dt + \int_0^{T/2} -\frac{1}{2} dt \right]$$

$$a_0 = \frac{1}{T} \left[-\frac{1}{2} |t|_{-T/2}^0 + \frac{1}{2} |t|_0^{T/2} \right]$$

$$\boxed{a_0 = 0}$$

$$a_n = \frac{2}{T} \int_{-T/2}^{T/2} f(t) \cdot \cos n\omega_0 t \cdot dt = 0$$

odd even

$$b_n = \frac{2}{T} \int_{-T/2}^{T/2} f(t) \cdot \sin n\omega_0 t \cdot dt$$

$$b_n = \frac{2}{T} \left[\int_{-T/2}^0 -\frac{1}{2} \cdot \sin n\omega_0 t \cdot dt + \int_0^{T/2} \frac{1}{2} \cdot \sin n\omega_0 t \cdot dt \right]$$

$$b_n = \frac{2}{T} \left[-\frac{1}{2} \left| -\frac{\cos n\omega_0 t}{n\omega_0} \right|_{-T/2}^0 + \frac{1}{2} \left| -\frac{\cos n\omega_0 t}{n\omega_0} \right|_0^{T/2} \right]$$

$$b_n = \frac{2}{T} \left[+\frac{1}{2} \left(\frac{1}{n\omega_0} - \frac{\cos n\omega_0 T/2}{n\omega_0} \right) - \frac{1}{2} \left(\frac{\cos n\omega_0 T/2}{n\omega_0} - \frac{1}{n\omega_0} \right) \right]$$

$$b_n = \frac{2}{T} \left[\frac{1}{2n\omega_0} - \frac{\cos n\omega_0 T/2}{2n\omega_0} - \frac{1}{2} \frac{\cos n\omega_0 T/2}{n\omega_0} + \frac{1}{2n\omega_0} \right]$$

$$b_n = \frac{2}{T} \left[\frac{1}{n\omega_0} - \frac{\cos n\omega_0 T/2}{n\omega_0} \right]$$

$$b_n = \frac{2}{Tn\omega_0} - \frac{2 \cos n\omega_0 T/2}{Tn\omega_0}$$

$$b_n = \frac{1}{n\pi} - \frac{\cos n\pi}{n\pi}$$

$$b_n = \frac{1}{n\pi} [1 - \cos n\pi]$$

$$n=1$$

$$n=2$$

$$n=3$$

$$b_1 = 2/\pi$$

$$b_2 = 0$$

$$b_3 = 2/3\pi$$

$n \rightarrow \text{even}$

$$b_n = 0$$

$n \rightarrow \text{odd}$

$$b_n \neq 0$$

$$\boxed{b_n = \frac{2}{n\pi}}$$

$$f(t) = \sum_{n=1}^{\infty} \left(\frac{2}{n\pi} \right) \sin n\omega_0 t$$

$n \rightarrow \text{odd}$

Polar form of Trigonometric Fourier Series

$$f(t) = a_0 + \sum_{n=1}^{\infty} (a_n \cos n\omega_0 t + b_n \sin n\omega_0 t)$$

$$a \cos \theta + b \sin \theta \quad \left| \quad \begin{aligned} a &= r \cos \phi \rightarrow a^2 = r^2 \cos^2 \phi \\ b &= r \sin \phi \rightarrow b^2 = r^2 \sin^2 \phi \end{aligned} \right. + \Rightarrow \begin{aligned} a^2 + b^2 &= r^2 \cos^2 \phi + r^2 \sin^2 \phi \\ a^2 + b^2 &= r^2 (\cos^2 \phi + \sin^2 \phi) \\ r^2 &= a^2 + b^2 \end{aligned}$$

$$r \cos \phi \cos \theta + r \sin \phi \sin \theta$$

$$r \cos \phi \cos \phi + r \sin \phi \sin \phi$$

$$r (\cos \phi \cos \phi + \sin \phi \sin \phi)$$

$$r \cos (\phi - \phi)$$

$$\longrightarrow r_0 \cos (\phi - \phi) = r_0 \cos \phi$$

$$f(t) = a_0 + \sum_{n=1}^{\infty} r_n \cos (n\omega_0 t - \phi_n)$$

$$r_n = \sqrt{a_n^2 + b_n^2}$$

$$\begin{aligned} b_n &= r_n \sin \phi_n \\ a_n &= r_n \cos \phi_n \end{aligned}$$

$$\frac{b_n}{a_n} = \tan \phi_n$$

$$f(t) = \sum_{n=0}^{\infty} r_n \cos (n\omega_0 t - \phi_n)$$

$$\phi_n = \tan^{-1} \left(\frac{b_n}{a_n} \right)$$

Problem on Phasor form

$$f(t) = 20 + \sin\left(\frac{\pi}{7}t + \frac{\pi}{3}\right) + 5 \sin\left(\frac{5\pi}{6}t - \frac{\pi}{4}\right) - \cos\left(\frac{6\pi}{5}t - \frac{\pi}{6}\right)$$

is it valid fourier series
find Harmonics?

$$f(t) = \sum_{n=0}^{\infty} r_n \cos(n\omega_0 t - \phi_n)$$

$$f(t) = 20 + \cos\left(\frac{\pi}{7}t + \frac{\pi}{3} - \frac{\pi}{2}\right) + 5 \cos\left(\frac{5\pi}{6}t - \frac{\pi}{4} - \frac{\pi}{2}\right)$$

$$= 20 + \cos\left(\frac{\pi}{7}t - \frac{\pi}{6}\right) + 5 \cos\left(\frac{5\pi}{6}t - \frac{3\pi}{4}\right) - \cos\left(\frac{6\pi}{5}t - \frac{\pi}{6}\right)$$

$$\omega_1 = \frac{\pi}{7}, \quad \omega_2 = \frac{5\pi}{6}, \quad \omega_3 = \frac{6\pi}{5}$$

$$\omega_0 = \frac{\text{HCF}(\omega_1, \omega_2, \omega_3)}{\text{LCF}(\omega_1, \omega_2, \omega_3)} = \frac{\text{HCF}(\pi, 5\pi, 6\pi)}{\text{LCM}(7, 6, 5)} = \frac{\pi}{210}$$

$$\omega_0 = \frac{\pi}{210}$$

$$K \times \frac{\pi}{210} = \frac{\pi}{7}$$

$$\boxed{K = 30} \checkmark$$

$$K \times \frac{\pi}{210} = \frac{5\pi}{6}$$

$$K = \frac{210 \times 5}{6}$$

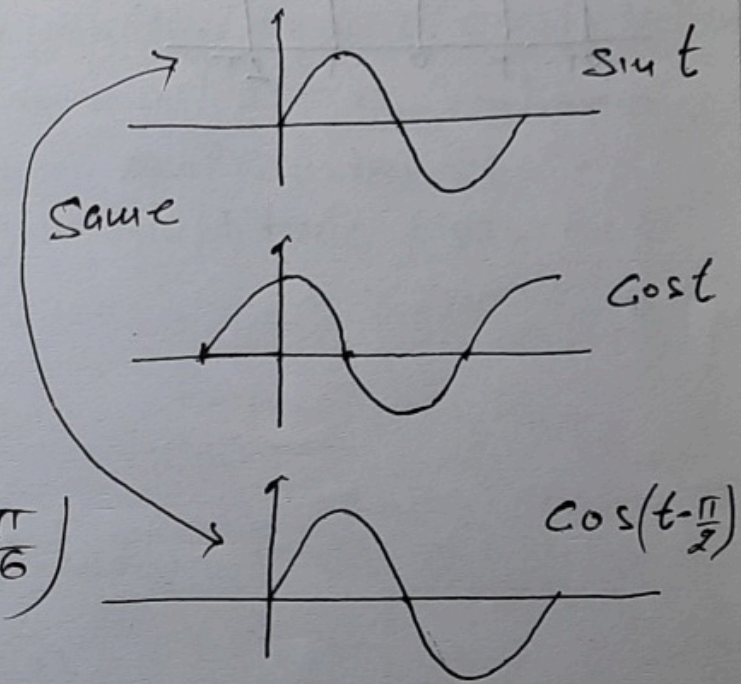
$$\boxed{K = 175}$$

$$K \times \frac{\pi}{210} = \frac{6\pi}{5}$$

$$K = \frac{210 \times 6}{5}$$

$$\boxed{K = 252}$$

BASIC $\downarrow$



Exponential Fourier Series

$$f(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t}$$

Where C_n is Exponential Fourier Co-efficient

$$C_n = \frac{1}{T} \int_{-T/2}^{T/2} f(t) \cdot e^{-jn\omega_0 t} dt$$

Relation b/w C_n & a_n, b_n

$$C_n = \frac{1}{T} \int_{-T/2}^{T/2} f(t) \cdot e^{-jn\omega_0 t} dt =$$

$$C_n = \frac{1}{T} \int_{-T/2}^{T/2} f(t) \cdot (\cos n\omega_0 t - j \sin n\omega_0 t) dt$$

$$C_n = \underbrace{\frac{1}{T} \int_{-T/2}^{T/2} f(t) \cdot \cos n\omega_0 t dt}_{\frac{a_n}{2}} - j \underbrace{\frac{1}{T} \int_{-T/2}^{T/2} f(t) \cdot \sin n\omega_0 t dt}_{\frac{b_n}{2}}$$

Why to study EFS:-

- * Easy method or easy computation
- * only to evaluate C_n unlike a_n, b_n in Real Fourier series.
- * Analyse Amplitude & Phase.

* Use this Property

$$e^{\pm j\phi} = \cos \phi \pm j \sin \phi$$
$$e^{-jn\omega_0 t} = \cos(n\omega_0 t) - j \sin(n\omega_0 t)$$

$$\Rightarrow \boxed{C_n = \frac{a_n}{2} - j \frac{b_n}{2}}$$

$$|C_n| = \sqrt{\left(\frac{a_n}{2}\right)^2 + \left(\frac{b_n}{2}\right)^2}$$

$$|C_n| = \frac{\sqrt{a_n^2 + b_n^2}}{2}$$

$$\boxed{|C_n| = \frac{\gamma_n}{2}}$$

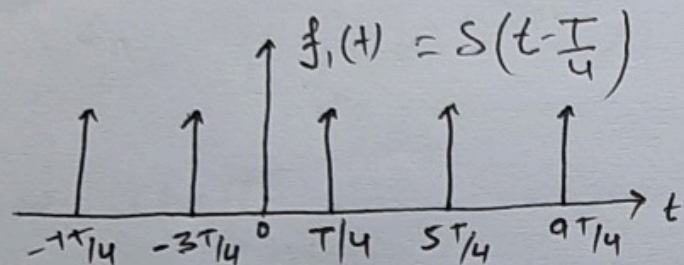
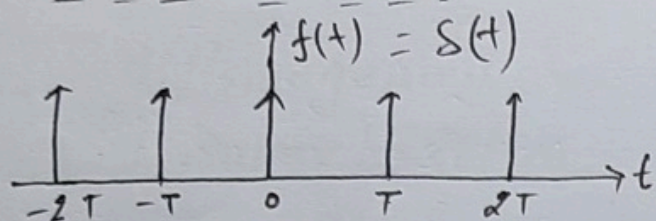
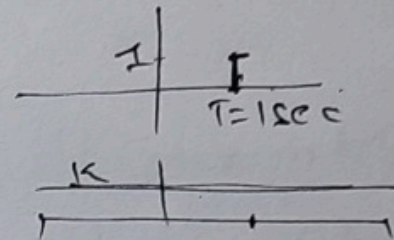
Properties of EFS

② Time shifting

$$C_n = \frac{1}{T} \int_{-\pi/2}^{\pi/2} (i/p) \cdot e^{-j n \omega_0 t} dt$$

$$C_n = \frac{1}{T} \int_{-\pi/2}^{\pi/2} \delta(t) \cdot e^{-j n \omega_0 t} dt$$

$$C_n = \frac{1}{T} \int_{-\pi/2}^{\pi/2} \delta(t) \cdot dt = \boxed{\frac{1}{T}} \checkmark$$



$$C_n = \frac{1}{T} \int_{-\pi/2}^{\pi/2} \underline{S(t - T/4)} e^{-j n \omega_0 t} dt$$

$$C_n = \frac{1}{T} \times e^{-j n \omega_0 T/4} \int_{-\pi/2}^{\pi/2} S(t - T/4) \cdot dt$$

$$C_n = \frac{e^{-j n \omega_0 T/4}}{T}$$

$$S(t) \xrightarrow{\text{EFC}} \frac{1}{T}$$

$$S(t - T/4) \xrightarrow{\text{EFC}} \frac{1}{T} \times e^{-j n \omega_0 T/4}$$

$$S(t + T/4) \xrightarrow{\text{EFC}} \frac{1}{T} \times e^{j n \omega_0 T/4}$$

$$f(t) \longrightarrow C_n$$

$$f(t - T_0) \longrightarrow C_n e^{-j n \omega_0 T_0}$$

$$f(t + T_0) \longrightarrow C_n e^{j n \omega_0 T_0}$$

Properties of Exponential F.S

② Linear Properties

i) Homogeneity

ii) Superposition

$$C_n = \frac{1}{T} \int_{-\pi/2}^{\pi/2} (i/p) \cdot e^{-j\omega_0 t} dt$$

$$i/p = a f_1(t)$$

$$C_n = \frac{1}{T} \int_{-\pi/2}^{\pi/2} a f_1(t) \cdot e^{-j\omega_0 t} dt$$

$$= a \times \frac{1}{T} \int_{-\pi/2}^{\pi/2} f_1(t) \cdot e^{-j\omega_0 t} dt$$

$$i/p = b f_2(t)$$

$$EFC = b C_{n2}$$

$$EFC = a C_{n1} + b C_{n2}$$

$$f_1(t) \rightarrow EFC \rightarrow C_{n1}$$

$$f_2(t) \rightarrow EFC \rightarrow C_{n2}$$

$$a f_1(t) \rightarrow EFC \rightarrow a C_{n1} \checkmark$$

$$b f_2(t) \rightarrow EFC \rightarrow b C_{n2}$$

$$a f_1(t) + b f_2(t) \rightarrow EFC \rightarrow a C_{n1} + b C_{n2}$$

$$i/p = a f_1(t) + b f_2(t)$$

$$EFC = \frac{1}{T} \int_{-\pi/2}^{\pi/2} (a f_1(t) + b f_2(t)) e^{-j\omega_0 t} dt$$

$$= \frac{1}{T} \int_{-\pi/2}^{\pi/2} (a f_1(t) e^{-j\omega_0 t} + b f_2(t) e^{-j\omega_0 t}) dt$$

$$= \underbrace{\frac{1}{T} \int_{-\pi/2}^{\pi/2} a f_1(t) e^{-j\omega_0 t} dt}_{a C_{n1}} + \underbrace{\frac{1}{T} \int_{-\pi/2}^{\pi/2} b f_2(t) e^{-j\omega_0 t} dt}_{b C_{n2}}$$

$$f_1(t) \rightarrow EFC \checkmark$$

$$f_2(t) \rightarrow EFC \checkmark$$

$$f_3(t) = f_1(t) + f_2(t) \\ \underline{\quad} = f_1(t) + f_2(t)$$

Properties of EFS

③ Time Differentiation

$$f(t) = \sum_{n=0}^{\infty} c_n \cdot \underline{e^{j n \omega_0 t}}$$

Exp Fourier Coeff.

diff $f(t)$ w.r.t time

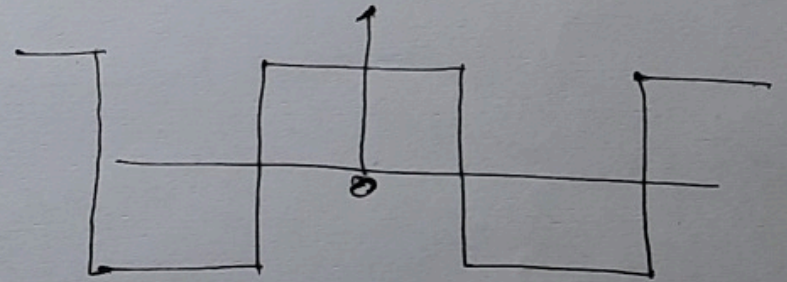
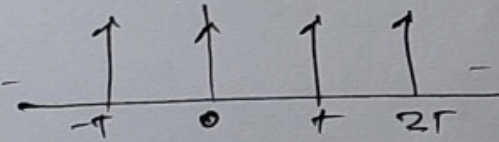
$$\frac{df(t)}{dt} = \sum_{n=0}^{\infty} c_n \frac{d}{dt} (e^{j n \omega_0 t})$$

$$\frac{df(t)}{dt} = \sum_{n=0}^{\infty} \underline{c_n \times j n \omega_0} e^{j n \omega_0 t}$$

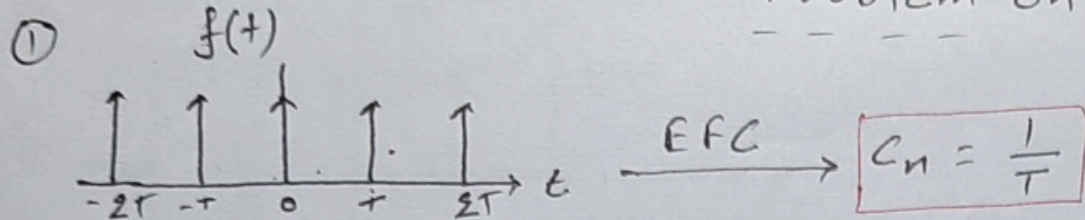
Exp Fourier Coeff.

$\left\{ \begin{array}{l} \\ \\ \\ \end{array} \right.$	$f(t) \longrightarrow c_n$
	$\frac{df}{dt} \longrightarrow c_n (j n \omega_0)$
	$\frac{d^k f}{dt^k} \longrightarrow c_n (j n \omega_0)^k$

$$e^{2x} \xrightarrow{\frac{d}{dx}} 2e^{2x}$$



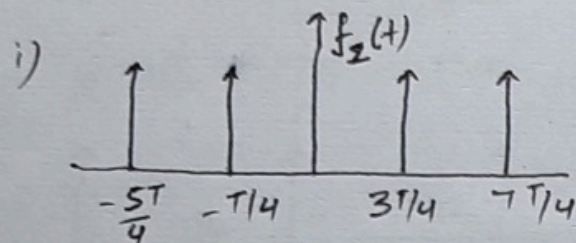
Problem on Properties



$$C_n = \frac{1}{T} \int_{-T/2}^{T/2} f(t) e^{-j n \omega_0 t} dt$$

$$C_n = \frac{1}{T} e^{j n \omega_0 (0)} \int_{-T/2}^{T/2} \delta(t) dt \quad \begin{matrix} \delta(t) = 0 & t \neq 0 \\ \delta(t) = 1 & t = 0 \end{matrix}$$

$$C_n = \frac{1}{T}$$



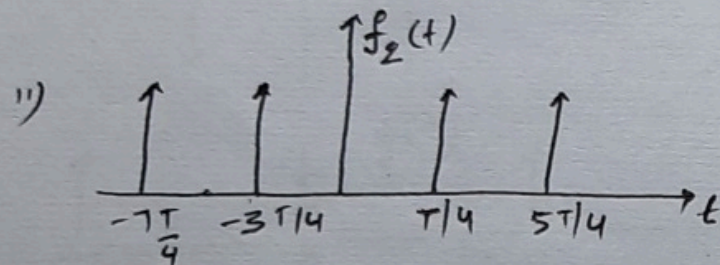
$$f_2(t) = f(t + T/4)$$

$$f(t) \rightarrow C_n$$

$$f(t + T/4) \rightarrow C_n \times e^{j n \omega_0 T/4}$$

$$f_1(t) \xrightarrow{\text{EFC}} \frac{1}{T} \times e^{j n \omega_0 T/4}$$

$$\boxed{\frac{1}{T} \times e^{j n \omega_0 T/4}}$$



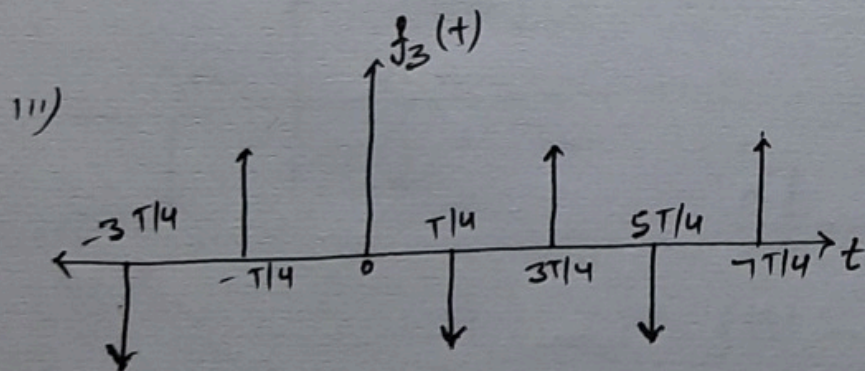
$$f_2(t) = f(t - T/4)$$

$$f(t - T/4) \rightarrow C_n e^{-j n \omega_0 T/4}$$

$$\xrightarrow{\text{EFC}} \boxed{\frac{1}{T} \times e^{-j n \omega_0 T/4}}$$

$$\frac{1}{T} \times 2J \sin\left(4 \frac{2\pi}{T} \times \frac{T}{4}\right)$$

$$\boxed{\frac{2J}{T} \sin\left(4 \frac{\pi}{2}\right)}$$



$$f_3(t) = f_1(t) - f_2(t)$$

$$f_3(t) \xrightarrow{\text{EFC}} \frac{1}{T} e^{j n \omega_0 T/4} - \frac{1}{T} e^{-j n \omega_0 T/4}$$

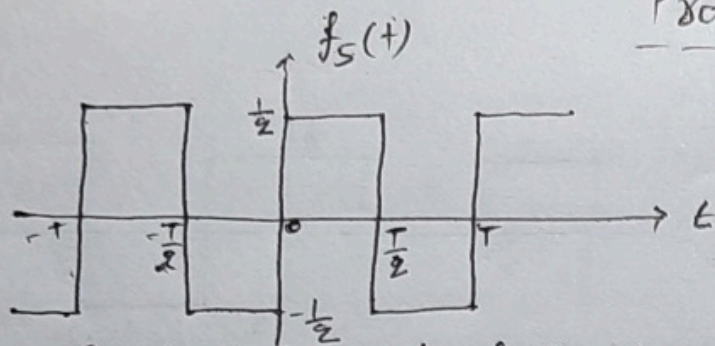
$$\frac{1}{T} \left[\frac{e^{j n \omega_0 T/4} - e^{-j n \omega_0 T/4}}{2J} \right] \times 2J$$

$$f_3(t) \xrightarrow{\text{EFC}} \frac{1}{T} \times 2J \sin(n \omega_0 T/4)$$

$$\left. \begin{matrix} \omega_0 = \frac{2\pi}{T} \\ T = \frac{2\pi}{\omega_0} \end{matrix} \right\} \frac{1}{T} \times 2J \sin(n \omega_0 T/4)$$

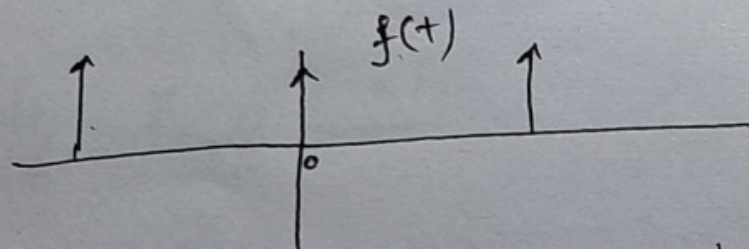
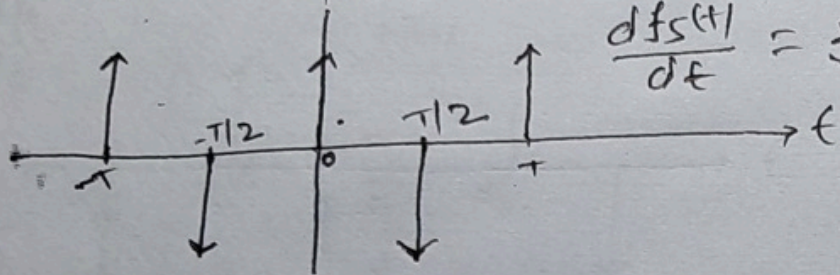
Problem on Properties

③

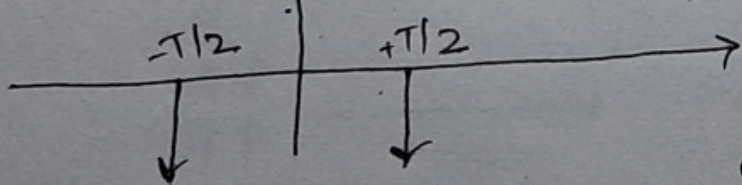


find Exponential Fourier Co-eff

$$\frac{df_s(t)}{dt} = f(t) - f(t-T/2)$$



$$-f(t-T/2)$$



$$f(t) + (-f(t-T/2))$$

$$f(t) - f(t-T/2)$$

$$\frac{df}{dt} \longrightarrow c_n$$

$$f(t) \longrightarrow \frac{c_n}{j n \omega_0}$$

$$f(t) \xrightarrow{\text{EFC}} \frac{1}{T}$$

$$-f(t-T/2) \longrightarrow -\frac{1}{T} e^{-j n \omega_0 T/2}$$

$$f(t) - f(t-T/2) \longrightarrow$$

$$\frac{1}{T} - \frac{1}{T} e^{-j n \omega_0 T/2}$$

$$\frac{1}{T} \left[1 - e^{-j n \omega_0 T/2} \right]$$

$$\frac{1}{T} \left[1 - \left(\cos n \omega_0 \frac{T}{2} - j \sin n \omega_0 \frac{T}{2} \right) \right]$$

$$\frac{1}{T} \left[1 - (\cos 4\pi - j \sin 4\pi) \right]$$

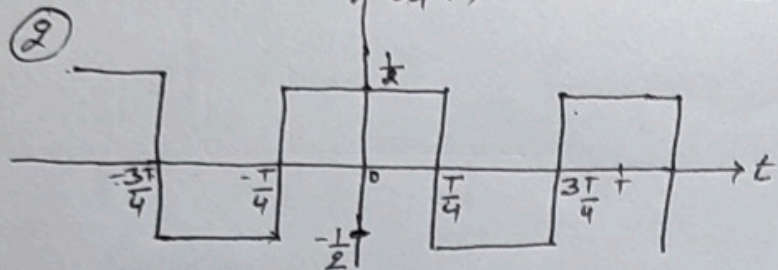
$$\frac{1}{T} [1 - \cos 4\pi]$$

$$f_s(t) \xrightarrow{\text{EFC}} \frac{1}{T} \times \frac{(1 - \cos 4\pi)}{j n \omega_0}$$

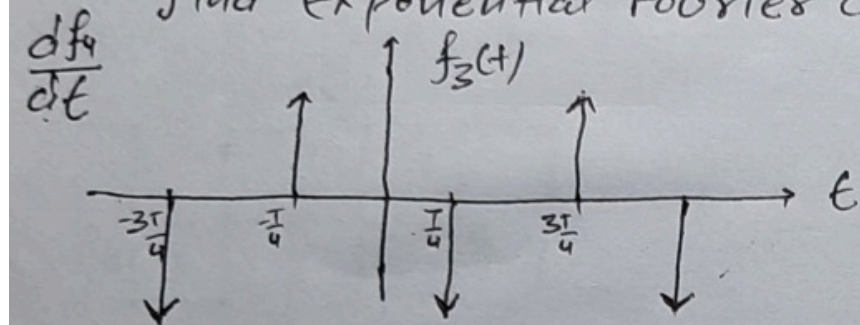
$$f_s(t) \xrightarrow{\text{EFC}} \boxed{\frac{1}{j n 2\pi} (1 - \cos 4\pi)}$$

Problem on Properties

$$x(t) \xrightarrow{d} u(t) \xrightarrow{d} \delta(t)$$



find Exponential Fourier Co-eff



$$\frac{df_4}{dt} = f_3(t) \xrightarrow{\text{EFC}} \frac{2J}{T} \sin\left(4\frac{\pi}{2}\right)$$

$$f(t) \xrightarrow{\text{EFC}} C_n$$

$$\frac{df}{dt} \longrightarrow C_n \times jn\omega_0$$

$$\frac{df_4}{dt} \xrightarrow{\text{EFC}} \frac{2J}{T} \times \sin\left(4\frac{\pi}{2}\right)$$

$$f_4(t) \xrightarrow{\text{EFC}} \frac{2J}{T} \times \frac{\sin\left(4\frac{\pi}{2}\right)}{jn\omega_0}$$

$$\frac{2J}{T} \times \frac{1 \times T}{jn 2\pi} \sin\left(4\frac{\pi}{2}\right)$$

$$\boxed{\frac{1}{jn\pi} \sin\left(4\frac{\pi}{2}\right)}$$

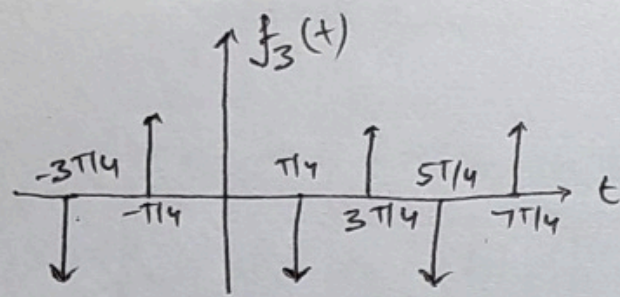
$$\int f(t) \longrightarrow C_n$$

$$\left[\frac{df}{dt} \longrightarrow C_n \times jn\omega_0 \right.$$

$$\int \frac{df}{dt} \longrightarrow C_n$$

$$\left[f(t) \longrightarrow \frac{C_n}{jn\omega_0} \right.$$

Important Conclusion



odd signal

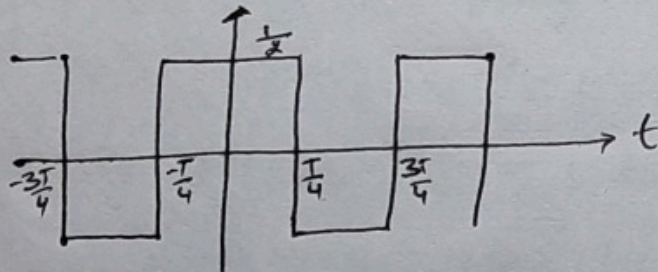
→ EFC

$$\frac{2J}{T} \sin\left(n\frac{\pi}{2}\right)$$

C_n imaginary value

$$C_n = \frac{a_n}{2} - j \frac{b_n}{2}, \quad a_n = 0, \quad b_n \neq 0$$

$n = 2$
even $\rightarrow n$



→ EFC

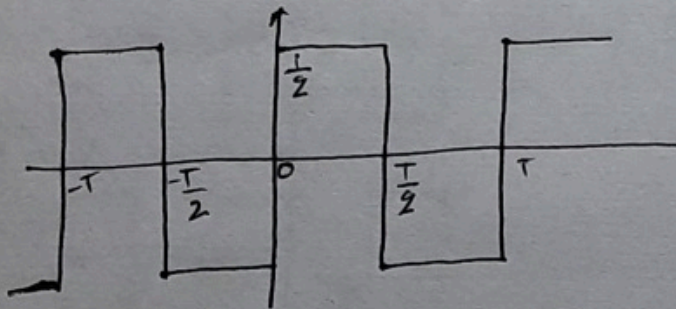
$$\frac{1}{n\pi} \sin\left(n\frac{\pi}{2}\right)$$

C_n Real value

$$C_n = \frac{a_n}{2} - j \frac{b_n}{2}, \quad a_n \neq 0, \quad b_n = 0$$

$n \rightarrow$ even
 $C_n = 0$

Real & Even signal



Real & odd signal

→ EFC

$$\frac{1}{jn2\pi} (1 - \cos n\pi)$$

C_n imaginary value

$$C_n = \frac{a_n}{2} - j \frac{b_n}{2}$$

$$a_n = 0$$

$$b_n \neq 0$$

$n \rightarrow$ even
 $C_n = 0$

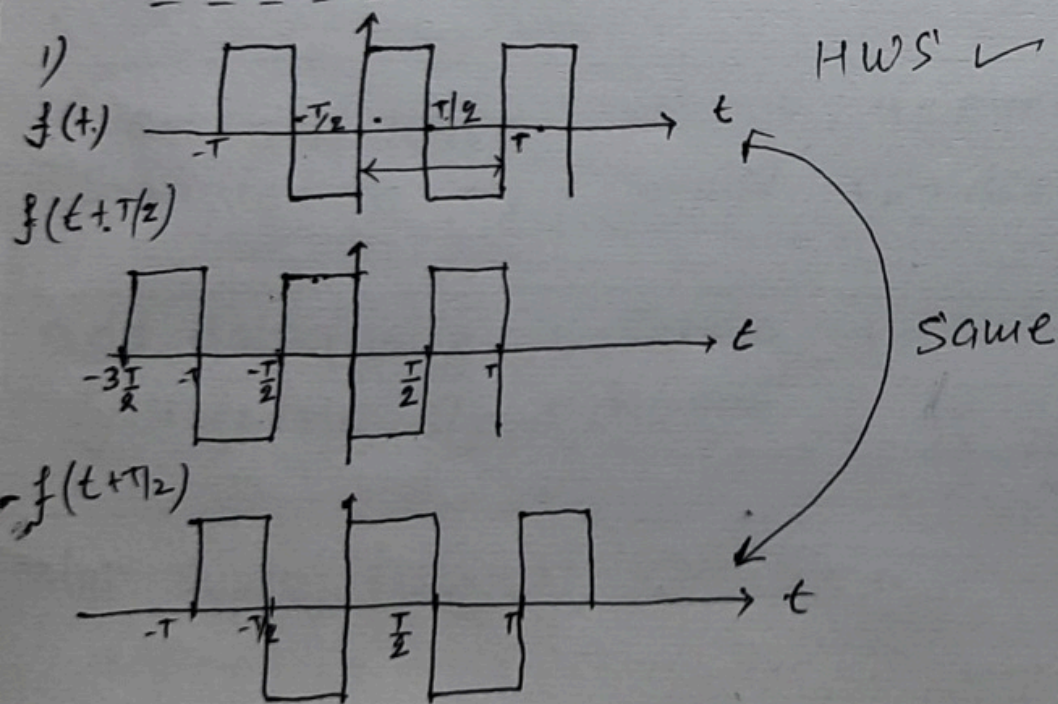
Half Wave Symmetry

* Condition $f(t) = -f(t \pm T/2)$

Procedure to check HWS

- i) let $f(t)$ given signal
- ii) Make time shifting operation ($f(t \pm T/2)$)
- iii) Make Amplitude Reversal ($-f(t \pm T/2)$)
- iv) if $f(t) = -f(t \pm T/2)$ then HWS otherwise not HWS

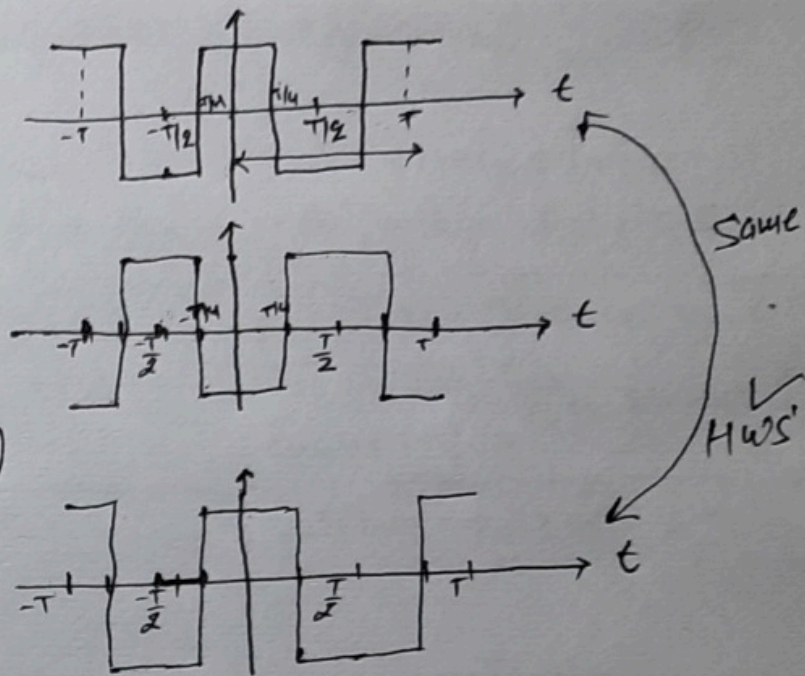
Problems



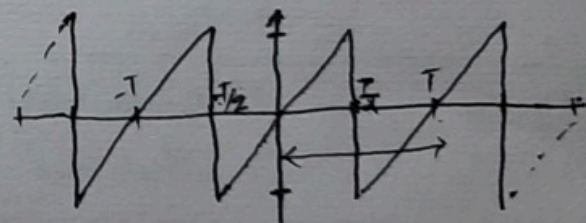
2) $f(t)$

$f(t - T/2)$

$-f(t - T/2)$



3) $f(t)$



Find it is HWS or Not

$f(t \pm T/2)$ ✓

$-f(t \pm T/2)$

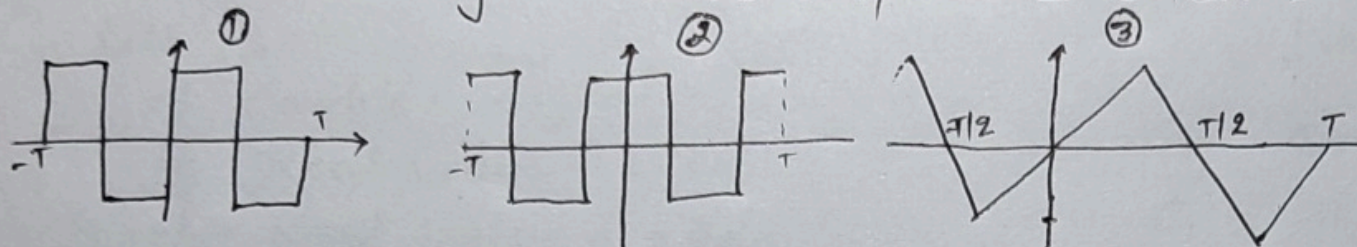
Homework

Most important Properties of Fourier Co-efficient

$f(t)$ Real signal	Exponential (C_n) Fourier co-efficient	Trigonometric Fourier coefficient	Trigonometric Fourier Series
Real & Even signal $f(t) = f(-t)$	Real $C_n = \frac{a_n}{2} - j \frac{b_n}{2}$	$a_0, a_n \neq 0$ $b_n = 0$	$f(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos n\omega_0 t$ [DC terms] + [Cosine] $\checkmark$
Real & odd signal $f(t) = -f(-t)$	Imaginary	$a_0, a_n = 0$ $b_n \neq 0$	Sine terms $\checkmark$
Half wave symmetry $f(t) = -f(t \pm T/2)$	$C_n = 0$ $n = \text{even}$	$a_0, a_n, b_n = 0$ $n = \text{even}$	odd Harmonics (sin, cos) $\checkmark$
Even Half wave Symmetric s/g	$C_n = 0, n = \text{even}$ $n = \text{odd} \rightarrow C_n \rightarrow \text{Real}$	$b_n = 0, \text{even, odd}$ $a_0, a_n \neq 0 \rightarrow \text{odd}$	odd Cosine terms
Odd Half wave Symmetric s/g	$C_n = 0 \rightarrow \text{even}$ $n \rightarrow \text{odd} \rightarrow C_n \rightarrow \text{Imaginary}$	$b_n \neq 0 \rightarrow \text{odd}$ $a_0, a_n = 0$	odd Sine terms
Non symmetric signal	Complex	$a_0, a_n \neq 0$ $b_n \neq 0$	—

Problems based on Properties

1) Which of the signals have only odd sine terms



$$f(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos n\omega_0 t + \sum_{n=1}^{\infty} b_n \sin n\omega_0 t$$

$$C_n = \frac{a_n}{2} - j \frac{b_n}{2}$$

odd sin terms

↓
HWS, $b_n \neq 0$
 $a_n = 0$

Ans signal 1 & 3

odd HWS

2) Which of the signals have only odd cosine terms

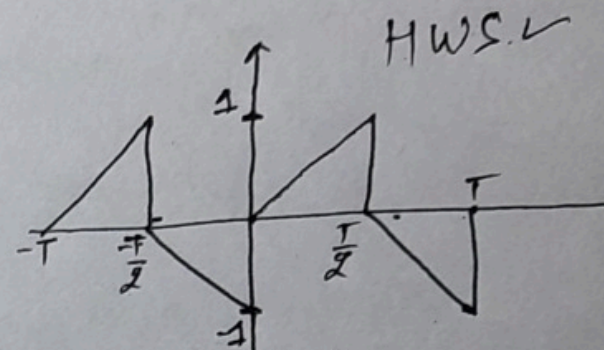
odd $\rightarrow$ HWS

only cosine $\rightarrow a_n \neq 0$
 $b_n = 0$

$\rightarrow C_n \rightarrow \text{Real}$

Ans $\rightarrow$ Signal 2

3)



This signal has

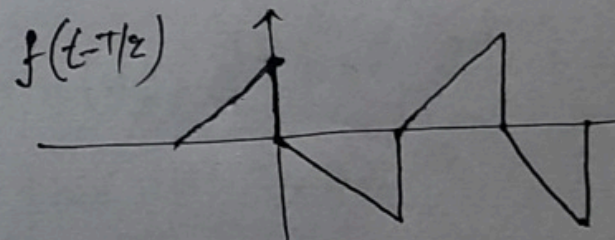
i) only sin terms

ii) only cosine terms

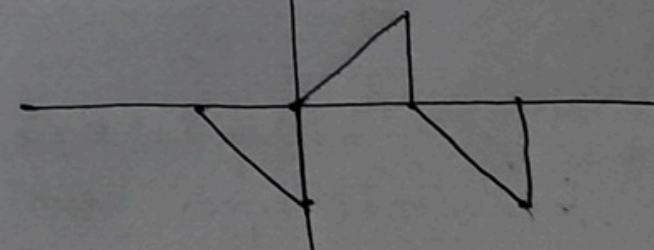
✓ iii) only odd Harmonics

iv) All of them

$$f(t) = -f(t \pm T/2)$$



$$-f(t - T/2) = f(t)$$



$$\textcircled{4} \quad f(t) = \sum_{n=0}^{100} \cos(n\pi) e^{jn \frac{2\pi}{100} t}$$

$f(t)$ is

- a) complex signal ✓
- b) Real value & even
- c) Real value & odd
- d) Real value but not even odd

$$f(t) = \sum_{n=-\infty}^{\infty} C_n e^{jn\omega_0 t}$$

$$C_n = \cos(n\pi)$$

$$n=1$$

$$C_1 = -1$$

$$C_{-1} = 0$$

$$C_{-1} = C_1^*$$

$$0 \neq -1$$

for Real signal

$$C_{-n} = C_n^*$$

$$C_n = \text{Real}$$

$$f(t) \rightarrow \text{Even}$$

$$C_n \rightarrow \text{Imaginary}$$

$$f(t) \rightarrow \text{odd}$$

$$\textcircled{5} \quad f(t) = \sum_{n=-100}^{100} \cos(n\pi) e^{jn \frac{2\pi}{100} t}$$

find the $f(t)$ type

$$C_n = \cos(n\pi) \rightarrow \text{Real number}$$

$$C_1 = -1$$

$$C_{-1} = \cos(-\pi) = -1$$

$$C_{-1} = C_1^* \rightarrow f(t) \rightarrow \text{Real signal}$$

$$\text{even signal}$$

$$[-1 = -1]$$

$$\textcircled{6} \quad f(t) = \sum_{n=-100}^{100} j \sin(n \frac{\pi}{2}) e^{jn \frac{2\pi}{100} t}$$

find $f(t)$ type

$$C_n = j \sin(n \frac{\pi}{2})$$

$$C_{-1} = j \sin(-\frac{\pi}{2})$$

$$C_1 = j \sin(\frac{\pi}{2})$$

$$= -j \sin(\frac{\pi}{2})$$

$$C_1 = j$$

$$C_{-1} = -j$$

$$C_{-1} = C_1^*$$

Real signal

odd signal

$$-j = j^*$$

Dirichlet's Condition

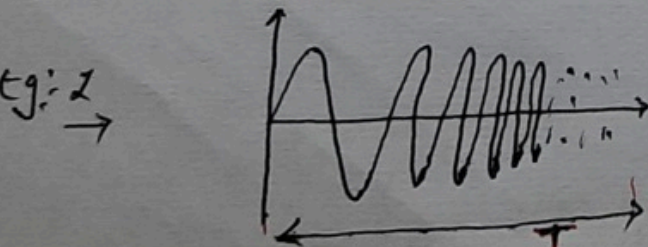
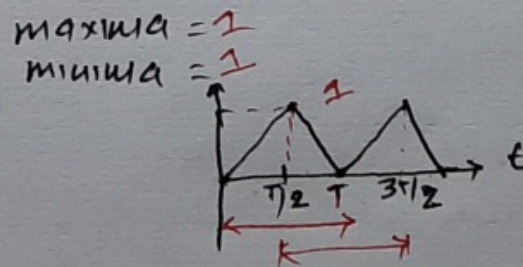
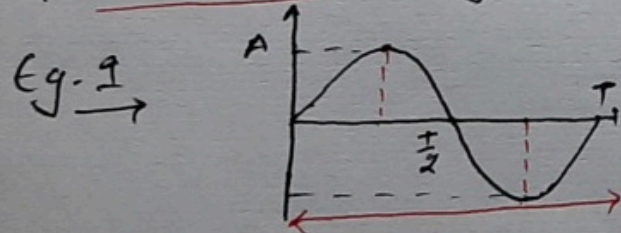
Basic

- * Periodic signal $\rightarrow$ Fourier Series
- * Aperiodic signal $\rightarrow$ Fourier Transform

- * But not all periodic signals $\rightarrow$ Fourier Series
- * If Periodic signal will follow 3 conditions then only it can represent in Fourier Series
- * These 3 conditions are known as Dirichlet Condition

Condition 1

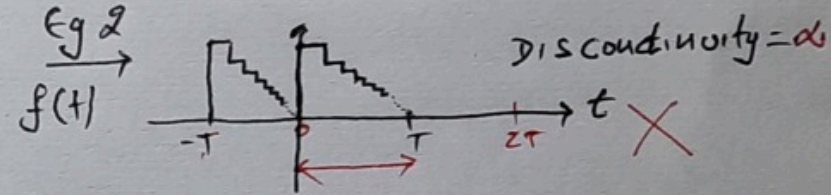
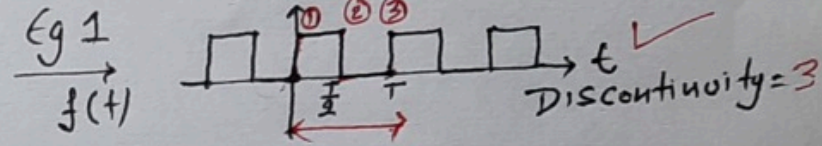
Periodic signal must have finite maxima & minima in given range of time Period



max = ∞
min = ∞

Condition 2

Periodic signal must have finite no of discontinuity over a given range of time Period



Condition 3

Periodic signal $f(t)$ must be absolutely integrable over the range of its time period. (T)
let T time period

$$\int_{-T/2}^{T/2} |f(t)| dt < \infty$$

